Ana Emilia de Orellana

https://anadeore.github.io
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EMPLOYMENT

Postdoctoral Researcher, University of Jyväskylä *Aug. 2026 - Present*
Funded by Tuomas Orponen's ERC Grant MUSING.

Junior Research Fellow, Institut Mittag-Leffler *Sep.–Dec. 2026*
Semester programme "Interactions between fractal geometry, harmonic analysis, and dynamical systems"

EDUCATION

School of Mathematics and Statistics, University of St Andrews *Sep. 2023 - May 2026*
PhD in Mathematics

Dissertation Title: *New Perspectives in Harmonic Analysis and Fractal Geometry.*
Supervisors: Prof. Jonathan M. Fraser, Regius Prof. Kenneth J. Falconer.

Facultad de Ingeniería Química, Universidad Nacional del Litoral *Feb. 2018 - Mar. 2023*
Licentiate Degree in Mathematics (equivalent to M.Sc.)
GPA: 8.68/10, part of the honour board.

Dissertation Title: *Homogeneous Type Spaces and Metrisation.*
Supervisors: Dr. Marisa Toschi, Dr. Mauricio Ramseyer.

PUBLICATIONS

- On Polynomial Progressions Inside Sets of Large Dimension, with B. Krause. 2026. arXiv: 2508.04680.
- Fourier restriction estimates based on L^q-dimensions: beyond Stein–Tomas, with M. Carnovale and J. M. Fraser. 2026. arXiv: 2606.07143.
- Knapp-type obstructions in multilinear fractal Fourier extension, with I. Oliveira. 2026. arXiv: 2602.08568.
- Fourier restriction for the additive Brownian sheet, with J. M. Fraser. 2026. arXiv: 2601.05802.
- L^2 restriction estimates from the Fourier spectrum, with M. Carnovale and J. M. Fraser. 2024. arXiv: 2412.14896.
- A Fourier analytic approach to exceptional set estimates for orthogonal projections, with J. M. Fraser. *Indiana Univ. Math. J.* **75** No. 3 (2026), pp. 679–707.
- Obtaining the Fourier spectrum via Fourier coefficients, with M. Carnovale and J. M. Fraser. *Proceedings of the American Mathematical Society* **154**(2026) pp. 2005–2017.

EVENTS ORGANISED

- Harmonic Analysis and PDEs Workshop at the University of St Andrews. *8th Apr. 2025*
  One day workshop for the LMS funded HAPDE Network.
  Co-organised with Jonathan Fraser.

· Pure PhD Seminar at the University of St Andrews.
Seminar series given by PhD students of the Analysis, Algebra, and Geometry research groups.
Co-organised with Violeta Lopez (Geometry PhD student). *Sep. 2025 - May 2026*

Analysis Reading Group

· Topics in Fractal Geometry, Harmonic Analysis, and Dynamical Systems. *Sep.–Dec. 2024*

AWARDS

· Nominee from the School of Mathematics and Statistics for the Watson-Arnott Awards for Research Excellence, in the category of Postgraduate Researcher Achievement. *Jul. 2026*

· St Leonard's College Scholarship, University of St Andrews. Funding to attend the 'Fractals and Related Fields V' at the Porquerolles Island, France. *Jun. 2026*

· St Leonard's College Scholarship, University of St Andrews. Funding to attend the 'Workshop on Ergodic Theory and Fractal Geometry' at the University of Loughborough. *Jul. 2024*

· Santander Development Fund Scholarship. Funding to attend the 'Geometry and Fractals under the Midnight Sun Conference' at the University of Oulu. *Jun. 2024*

· University of St Andrews - Handsel Scholarship for PhD in Mathematics. *Sep. 2023 - Mar. 2027*

· Second place in the National Monograph competition held by the Argentinian Mathematical Union. Title: 'Knot Theory'. *Sep. 2019*

EXPERIENCE

M.Sc. and Licenciate Degree examination *Aug. 2026*

Thomas Agustín Martín (University of Mar del Plata, Argentina). Interactions Between Fourier Analysis and Geometric Measure Theory: The Fourier Dimension Spectrum.

Academic Tutor and Demonstrator *Sep. 2023 - May. 2026*

Tutor at the University of St Andrews for a variety of pure and applied undergraduate mathematics modules. Marking summative and formative assessment for over 290 students. Demonstrator for over 650 students in examples classes.

Lecturer *Feb. 2023*

Lecturer at the Universidad Nacional del Litoral's 'Summer School' for entrance level students studying core mathematics modules. Lectured two groups of 70 students twice per week in two-hour sessions.

Teaching Assistant *Aug. 2022 - Mar. 2023*

Teaching assistant at Universidad Nacional del Litoral for first-year level undergraduate students.

Educational outreach *Mar. 2017 - Dec. 2018*

Collaborating with the Universidad Nacional del Litoral, providing one-to-one tuition in a variety of STEM subjects to less affluent members of the community.

TALKS

· Finding 3APs in fractal sets.
Interactions between fractal geometry, harmonic analysis, and dynamical systems, Institut Mittag-Leffler, Sweden. *Sep. 2026*

· Fourier restriction and a continuum of dimensions.
Fractals and Related Fields V, Porquerolles Island, France. *Jun. 2026*

- Multifractal measures and the Fourier restriction problem.
  Geometric measure theory and Metric geometry, University of Warwick, UK. *Mar. 2026*
- Fourier Restriction and Fractal Dimensions.
  Workshop on Regularity Theory of PDEs and Calculus of Variations, University of Essex, UK. *Jun. 2025*
- Extending the Stein–Tomas Theorem.
  Workshop in Harmonic Analysis and PDEs, University of St Andrews, UK. *Apr. 2025*
- Dimension Interpolation and Restriction Theorems.
  Young Functional Analysts' Workshop, University of Glasgow, UK. *Apr. 2025*
- Fourier Restriction for Multifractals.
  Ergodic Theory and Dynamical Systems Seminar, University of Bristol, UK. *Mar. 2025*
- A New Perspective on the Stein–Tomas Theorem.
  One World Fractals (online). *Nov. 2024*
- Fourier Analytic Methods for Orthogonal Projections.
  Analysis Seminar, Shenzhen Technology University, China (online talk). *Oct. 2024*
- Orthogonal Projections and a Continuum of Dimensions.
  Analysis Seminar, University of Birmingham, UK. *Oct. 2024*
- Dimension Interpolation and Exceptional Set Estimates.
  Analysis Seminar, University of Edinburgh, UK. *Sep. 2024*
- The Continuity of Exceptional Estimates for Orthogonal Projections.
  Fractal Geometry and Stochastics 7, Chemnitz, Germany. *Sep. 2024*
- Orthogonal Projections and the Fourier Spectrum.
  Workshop on Fourier Analysis, Fractals, and Finite Fields, University of St Andrews, UK. *Sep. 2024*
- Projection Theorems for the Fourier Spectrum.
  Workshop on the Geometry of Deterministic and Random Fractals II, Rényi Institute, Budapest, Hungary. *Sep. 2024*
- Fourier Analysis for Fractal Measures.
  Workshop on Ergodic Theory and Fractal Geometry, University of Loughborough, UK. *Jul. 2024*
- Exceptional Projections and Dimension Interpolation.
  Geometry and Fractals under the Midnight Sun Conference, University of Oulu, Finland. *Jun. 2024*

REFERENCES

- Prof. Jonathan Fraser: jmf32@st-andrews.ac.uk